When Does Training-Free DTW Match Learned Models on Hand Landmarks? A Data-Efficiency and Signer-Independence Study of Isolated Sign Recognition

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Abstract

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1 Introduction

Isolated sign recognition assigns a label from a fixed vocabulary to a short video of one sign. It is a small part of sign language processing and is not translation: it ignores grammar, context and most non-manual signals. We study it because it is a clean setting in which to ask a practical methods question. When there are only a few labelled examples per sign, does matching a new video against stored examples with dynamic time warping (DTW) work as well as a trained neural network, and at what point does the network pull ahead?

A common recipe extracts hand and body landmarks with MediaPipe (Lugaresi et al., 2019; Zhang et al., 2020), turns them into per-frame features, and either trains a sequence model or compares sequences directly. Nearest-neighbour DTW needs no training and can be updated by adding examples, which is attractive for small, changing vocabularies. In general time-series classification, nearest-neighbour DTW with a tuned warping window is a strong baseline (Bagnall and Lines, 2014), and deep networks are competitive on large benchmarks (Ismail Fawaz et al., 2019). We have not found a controlled comparison of the two on signer-independent sign data that varies both the number of examples per sign and the vocabulary size, so we ran one.

What this paper does. On PopSign ASL v1.0 (Starner et al., 2023), with the dataset's signer-disjoint split, we compare DTW nearest-neighbour matching, DTW prototypes, Euclidean nearest neighbour, a 1D-CNN, a BiGRU and a small Transformer, all on the same MediaPipe features and the same queries. We vary the references per sign from 1 to all, the vocabulary from 10 to ?? signs, the feature set, the handling of missing hands and the DTW variant. The analysis was written down before the test data were scored (`PREREG.md`), and uncertainty is estimated by resampling signers.

What it does not do. It does not propose a new method, claim state-of-the-art accuracy, or show that the systems help Deaf people. It covers one sign language (American Sign Language), one dataset, smartphone selfie video, and a mostly one-handed vocabulary. The work was done by hearing researchers without a Deaf co-author; Section 7 discusses what that limits.

Relation to earlier work by the same author. An earlier conference paper (Mankar et al., 2024) described a MediaPipe and DTW sign recogniser built on an open-source implementation (gabguerin, 2022). That paper compared DTW with an LSTM on a small custom dataset without signer-independent evaluation. The present study keeps that pipeline's feature idea as one baseline and replaces its evaluation.

2 Related work

Datasets. Isolated-sign datasets include WLASL (Li et al., 2020) (over 2000 words, over 100 signers) and MS-ASL (Joze and Koller, 2018) (1000 signs, over 200 signers); ASL Citizen (Desai et al., 2023) (2731 signs, 52 signers), collected from consenting contributors; PopSign (Starner et al., 2023), recorded on smartphones; LSA64 (Ronchetti et al., 2023) for Argentinian Sign Language; and LSFB-ISOL (Fink et al., 2021) for Belgian French Sign Language. We use PopSign because its licence permits derived data, its signers consented to a public database, and it ships a signer-disjoint split.

Landmark-based recognition. Skeleton and pose-based models include graph convolutional

networks (Jiang et al., 2021) and a pose-based Transformer (Bohacek and Hruz, 2022). These are trained models evaluated on their own benchmarks; our smaller models are not meant to reproduce them, only to represent the learned side of a data-efficiency comparison.

DTW and template matching for signs. DTW (Sakoe and Chiba, 1978) has several variants: banded and approximate forms (Salvador and Chan, 2007), averaged prototypes (DBA, Petitjean et al., 2011) and the differentiable soft-DTW (Cuturi and Blondel, 2017). For signs, Arcanjo et al. (2024) apply FastDTW to angular features computed from MediaPipe landmarks, and Saggio et al. (2020) compare k-NN with DTW against a CNN on sensor-glove data. Vandendriessche et al. (2025) study one-shot recognition against a sign dictionary with learned embeddings. We did not find prior work that varies references per sign and vocabulary size together on signer-independent video data; our search was not exhaustive. The features we call F1 follow the hand-angle vectors of the open-source implementation (gabguerin, 2022) used in our earlier work (Mankar et al., 2024).

Ethics of sign language technology. Interdisciplinary and Deaf-led work argues that sign language AI has often been shaped by hearing researchers' priorities, rests on unrepresentative datasets and overstates communication benefits (Bragg et al., 2019, 2021; Desai et al., 2024). A survey of Deaf community members reports their views on automatic translation (Tran et al., 2023). We take these as constraints on how we scope and word this paper.

3 Data and methods

Data. We use the `game` category of PopSign ASL v1.0 (Starner et al., 2023): 250 signs recorded on smartphones, CC BY 4.0, with a training split of 31 signers, a validation split of 8 and a test split of 8. The code checks that no signer appears in two splits. Vocabularies of 10, 50, 100 and 250 signs are nested prefixes of a fixed random order of the signs. A vocabulary is reported only if every one of its signs was processed; others are marked as not run. Videos are downloaded, converted to landmarks and deleted.

Landmarks and features. We use MediaPipe Holistic 0.10.14 (Lugaresi et al., 2019) on frames resized to a 480 pixel short side. Coordinates are scaled by frame width and height before any angle is computed (the earlier implementation used normalised coordinates, which distorts angles on portrait frames). A sequence is trimmed to the span between the first and last frame with a detected hand. Feature set F1 is the 210 pairwise angles between the 21 hand connection vectors per hand (identical information to the earlier 21×21 angle matrix; Euclidean distances differ by a constant factor). F2 is wrist-centred, scale-normalised hand shape. F3 combines F1 and F2, F4 adds hand location relative to the shoulders, F5 adds upper-body pose and F6 adds face-contour landmarks. The main grid uses F1. Missing hands are zero-filled in the main setting.

Methods. **DTW 1-NN** compares a query with every stored reference using DTW with Euclidean local cost and a Sakoe–Chiba band chosen on validation signers, and returns the label of the nearest reference. **Euclidean 1-NN** compares sequences resampled to 32 frames. **DBA prototype** replaces each sign's references by their DTW barycentre. The learned models are a 1D-CNN, a two-layer BiGRU and a three-layer Transformer encoder (each about 0.5M parameters) trained with cross-entropy for a fixed number of gradient steps with no early stopping, so no validation labels enter training. DTW distances are computed by our own compiled implementation, which is tested against tslearn (Tavenard et al., 2020). Learned-model learning rate and step budget are chosen on validation signers from a four-point grid per model and then fixed.

Protocol. References (or training examples) per sign $k \in \{1, 2, 5, 10, 20, \text{all}\}$ are drawn at random from the training split, nested across k, with three draw seeds (which also seed network initialisation). Test queries are up to three videos per test signer and sign, drawn once and shared by all methods. A query on which no hand is ever detected is counted as an error. We report top-1 accuracy with 95% intervals from resampling the 8 test signers (10,000 resamples, percentile intervals; cf. Field and Welsh, 2007), and paired differences use the same resamples. With only 8 clusters the intervals are wide and we do not report p-values. Two hypotheses were fixed in advance and judged at 97.5% (§4); all other comparisons are exploratory.

4 Results

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5 Discussion

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6 Limitations

- **One dataset, one language.** All results are for ASL on PopSign (47 signers, selfie phone video, mostly one-handed signs). We do not test other datasets, languages, cameras or continuous signing.
- **Few signers.** The test split has 8 signers, so signer-clustered intervals are wide and small differences are not resolvable. Training noise (3 seeds) is smaller than signer variation.
- **Small models, limited tuning.** The learned models are small and tuned on a four-point grid at one condition (vocabulary 50, 10 examples per sign). Larger models, pre-training or augmentation may change where learned models overtake DTW. We include one augmentation arm to test this at one setting only.
- **Landmark quality.** Results depend on MediaPipe 0.10.14 and on the resize used; the fraction of videos or frames without detected hands is reported in the tables. Face and other non-manual signals are mostly not modelled.
- **Not translation, not an application.** Top-1 accuracy on isolated signs is not communication access. We did not evaluate any system with Deaf users.
- **No phonological analysis.** PopSign has no phonological annotation; we report data-driven confusions and make no claim about phonological similarity.
- **Timing.** Latency was measured on one Apple M4 Max laptop with other processes running; no low-power device was available, so embedded performance is not measured.
- **Pre-registration scope.** The plan was written by the author before test scoring, but is not a registered report and was not independently time-stamped; the git history is the record.

7 Ethics and broader impact

Who did the work. This study was designed and run by hearing researchers without a Deaf co-author. Deaf-led analysis of sign language AI argues that such work often centres hearing researchers' priorities and a "communication barrier" framing (Desai et al., 2024). We chose a methods comparison on an existing, consented dataset rather than an application, and we make no claim that any result benefits Deaf people.

Data. PopSign participants consented to inclusion in a public database, and the dataset card lists identification and appearance-based uses as out of scope; we keep face landmarks local, release only hand and pose landmarks, and never attempt to identify signers. The dataset card also discourages continuous-signing and translation claims, which we avoid.

Risks. A cheap recogniser could be presented as a substitute for interpreters or language access. High accuracy on 250 isolated signs does not support that, and we discourage it. Community perspectives on automatic sign language translation (Tran et al., 2023) and dataset governance (Bragg et al., 2021) should be consulted before any deployment.

Community feedback. [To be completed by the author before submission: state whether Deaf community members reviewed the framing of this paper, what changed as a result, or that no such review took place.]

8 Reproducibility

Code, configuration, pinned requirements, per-query results and generated tables are in the accompanying repository (`make quick` reruns everything; `make report` regenerates every table, figure and number in this paper from the stored results). Landmark extraction versions and timings are logged, and every result file has a provenance file with the commit and library versions. The pre-registration is `PREREG.md`; deviations from it are listed there. Derived hand and pose landmarks are released under CC BY 4.0 with attribution to Georgia Tech and the Deaf Professional Arts Network; videos and face landmarks are not released.

Use of AI tools. An AI coding assistant (Claude, Anthropic) helped write code, tests and a first draft of the text and checked references. The author

specified the design, reviewed and ran the code, checked every number against the generated results files and every citation against its publisher or arXiv record, and takes responsibility for the content.

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